SPECTRAL ARITHMETIC SURGERY: PINNED FULL–MINOR ASSIGNMENT, ALL–RANK ARITHMETIC OBSTRUCTIONS, AND EXACT SALEM HEIGHT SURGERY

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Abstract. Let T be a rational matrix self-adjoint for a positive rational form H, let $e = e_1$ be cyclic, and prescribe simultaneously the characteristic polynomial of T and that of its first principal compression. We develop an exact arithmetic surgery calculus for low-rank H-self-adjoint perturbations and several rank-free obstructions tied to a fixed rational symmetrizing form. In rank one, a prescribed rational full–minor pair is attainable if and only if its cross polynomial is a scalar square of a mixed cofactor. This yields a finite cyclic-coset integrality test and a sharp quartic Salem obstruction for every $K \geq 5$, with an exception at $K = 4$. In rank two we derive the exact Gram–Plücker law, including its exterior correction, prove the converse assignment theorem, and reduce every fixed two-plane problem to rational linear elimination and one quadric.

For the canonical order-three realization we prove two all-rank barriers. Semidefinite output-pin surgery collapses to rank one, while arbitrary rational surgery preserving the canonical form must satisfy an explicit binary norm equation. A local obstruction to this equation produces an infinite family of genuine height-one Salem targets unattainable at every rank. On the canonical rank-two plane we strengthen the exact square-class criterion to a rational parametrization and finite integer-square sieve and show that every strict branch is necessarily indefinite. The unique integral target capable of a vanishing exterior shift is birational to the rank-zero elliptic curve of Cremona label $240a3$; its known rational points, also recovered by a short 2-isogeny descent, exclude every $K \geq 5$. In contrast, a sporadic $K = 5$ certificate and an infinite family $K = \nu^2 + 1$ give exact indefinite rank-two height collapse. These results close several natural surgery routes but do not prove a Salem-number closure conjecture.

1. Introduction

The characteristic polynomial of a rank-one perturbation is governed by the matrix determinant lemma. The inverse problem becomes subtler when one asks to prescribe, at the same time, the characteristic polynomial of a fixed principal compression. It becomes arithmetic when both outputs must be integral and geometrically constrained by interlacing. The purpose of this

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paper is to isolate the exact algebraic image of this simultaneous operation in ranks one and two and to test it on a rigid family of Salem pencils.

The classical ingredients—adjugates, resolvents, cyclic vectors, low-rank determinant formulas, restricted-rank spectral modification, principal-minor relations, and interlacing—are standard; see, for example, [8, 13, 14, 11, 2, 1, 7]. The contribution here is their simultaneous full–first-principal-compression arithmetic packaging for one fixed cyclic model and the resulting exact classification and obstruction theorems. The Salem quartics used below are classical specializations of Boyd's criterion [4, 16]; no novelty is claimed for the polynomials themselves.

Here is the main chain of results.

(i) A rational output pair (f_*, P_*) is produced by a fixed cyclic rank-one model if and only if

$$fP_* - f_*P = tA^2$$

for an explicitly reconstructible mixed cofactor A.

(ii) Two output pins force $X^2(X-2)^2$ to divide this cross polynomial. For a fixed pin direction, simultaneous integrality is membership in one finite cyclic subgroup of $\mathbf{Q}[X]/\mathbf{Z}[X]$.

(iii) The quartics

$$z^4 - (K-1)z^3 + z^2 - (K-1)z + 1$$

have exact normal height K, but for $K \geq 5$ their canonical pencils admit no nonzero signed two-pin rank-one surgery with integral Salem output.

(iv) In rank two the square is replaced by a binary Gram term minus an exterior correction. This identity again has a converse, hence it is an exact assignment criterion rather than merely a necessary condition. For a fixed update plane, feasibility reduces further to rational linear elimination and one quadric.

(v) Fixing the full and compressed eigenvectors is much stronger than fixing the output roots. An effective rank-r update fixing a full eigenline and a d-dimensional compressed eigenspace satisfies $r \leq N-1-d$. In the strict height-one slice, semidefinite output pins force these eigenspace pins, so every-rank semidefinite surgery is excluded for $K \geq 5$.

(vi) For the canonical rational symmetrizing form, every rational target of any rank satisfies an explicit binary norm equation. Primes $p \equiv 5, 7 \pmod 8$ give an infinite family of height-one Salem targets for which this equation is locally insoluble.

(vii) For the plane generated by $T_K e, T_K^2 e$, the complete rational image of the height-one target slice is a conic. We parametrize it, obtain a finite integer-square sieve at fixed K, and prove that every strict branch is indefinite.

(viii) The sole integral target capable of a vanishing exterior shift is birational to the elliptic curve $240a3$. Its eight rational points exclude all $K \geq 5$ and imply a uniform asymptotic gap of at least $1/2$ from the base exterior trace throughout the attainable integral slice.

(ix) In the opposite direction, the slice contains a sporadic $K = 5$ certificate and an infinite integral Salem family with base height $K = \nu^2 + 1$ and output height one. None of these outputs occurs in rank one.

The positive and negative results delimit one concrete surgery program. Height collapse is possible, but in the complete slice it is necessarily indefinite; the semidefinite, fixed-eigenspace, same-form local-obstruction, and canonical vanishing-shift routes are closed. Changing the symmetrizing form, changing the update plane, or moving to higher order remains outside these obstructions. In particular, no statement below proves a Boyd closure conjecture.

2. Cyclic self-adjoint pencils and cofactors

Throughout, unless stated otherwise,

$$T \in M_N(\mathbf{Q}), \qquad H = \operatorname{diag}(1, H') \in M_N(\mathbf{Q}), \quad H > 0, \qquad HT = T^{\mathsf{T}} H. \tag{1}$$

Put $e = e_1$, let $B = T^{(1)}$ be the first principal compression, and set

$$f = \det(XI - T), \qquad P = \det(XI - B). \tag{2}$$

We assume that e is cyclic for T and, when interlacing is invoked, that f and P are separable and coprime. Positive H-self-adjointness then places the problem in the usual real self-adjoint category after conjugating by $H^{1/2}$.

Lemma 2.1 (Rational symmetrizable realization)**.** *Let $f, P \in \mathbf{Q}[X]$ be monic of degrees $N, N-1$, respectively, with simple real zeros in strict interlacing position. Then there is a rational tridiagonal matrix*

$$T = \begin{pmatrix} a_0 & 1 & & \\ b_1 & a_1 & \ddots & \\ & \ddots & \ddots & 1 \\ & & b_{N-1} & a_{N-1} \end{pmatrix}, \qquad b_i \in \mathbf{Q}_{>0},$$

whose characteristic pair is (f, P). It is self-adjoint for

$$H = \operatorname{diag}\bigl(1, b_1^{-1}, (b_1 b_2)^{-1}, \dots, (b_1 \cdots b_{N-1})^{-1}\bigr),$$

and e_1 is cyclic.

Proof. The rational function P/f has simple positive residues at the zeros of f. Its finite Stieltjes–Jacobi continued fraction is obtained by the Euclidean algorithm over $\mathbf{Q}[X]$. Strict interlacing makes every off-diagonal product b_i positive, while monicity normalizes each superdiagonal entry to one. The continuant recurrence gives the stated characteristic pair. The displayed

diagonal form then satisfies $HT = T^{\mathsf{T}}H$, and nonzero off-diagonal products make e_1 cyclic; compare [10, 9]. □

For $u = (u_1, u') \in \mathbf{Q}^N$ define

$$A_u(X) = e^{\mathsf{T}} \operatorname{adj}(XI - T)u, \tag{3}$$

$$C_u(X) = u^{\mathsf{T}} H \operatorname{adj}(XI - T)u, \tag{4}$$

$$D_u(X) = (u')^{\mathsf{T}} H' \operatorname{adj}(XI - B)u'. \tag{5}$$

Lemma 2.2 (Cyclic cofactor normal form)**.** *For every $u \in \mathbf{Q}^N$ there is a unique $p \in \mathbf{Q}[X]$, $\deg p < N$, with $u = p(T)e$. For this polynomial,*

$$A_u = \operatorname{rem}_f(pP), \qquad C_u = \operatorname{rem}_f(p^2P), \qquad PC_u - A_u^2 = fD_u. \tag{6}$$

Proof. Cyclicity gives the unique representation. If λ is a zero of f, the simple-root adjugate formula, or equivalently spectral resolution of the resolvent, gives

$$A_u(\lambda) = p(\lambda)P(\lambda), \qquad C_u(\lambda) = p(\lambda)^2 P(\lambda).$$

The degree bounds yield the two remainder formulas. Write $G = (XI - T)^{-1}$. Self-adjointness gives

$$e^{\mathsf{T}}Gu = u^{\mathsf{T}}HGe = A_u/f, \qquad u^{\mathsf{T}}HGu = C_u/f.$$

Apply Sherman–Morrison to $T + tuu^{\mathsf{T}}H$. Comparing the $(1,1)$ resolvent entry, which equals the compression determinant divided by the full determinant, gives

$$\frac{P_t}{f_t} = \frac{P}{f} + \frac{t(A_u/f)^2}{1 - tC_u/f}.$$

Together with $f_t = f - tC_u$ this is precisely $P_t = P - t(PC_u - A_u^2)/f$. The second determinant lemma applied to the compression identifies the quotient with D_u. □

3. The rank-one square law and its converse

For $t \in \mathbf{Q}$ consider the signed H-self-adjoint update

$$T_t = T + tuu^{\mathsf{T}}H. \tag{7}$$

The determinant lemma and Lemma 2.2 give

$$f_t = f - tC_u, \qquad P_t = P - tD_u, \qquad \boxed{fP_t - f_tP = tA_u^2}. \tag{8}$$

Theorem 3.1 (Exact rank-one full–minor assignment)**.** *Fix the cyclic realization (1)–(2), and let f_*, P_* be monic rational polynomials of degrees $N, N-1$. For a fixed $t \in \mathbf{Q}^\times$, the following are equivalent.*

(a) *There is $u \in \mathbf{Q}^N$ such that $T + tuu^{\mathsf{T}}H$ has full and first compressed characteristic polynomials f_* and P_*.*

(b) *There is $A \in \mathbf{Q}[X]$, $\deg A < N$, such that*

$$fP_* - f_*P = tA^2. \tag{9}$$

In case (b)*, the update vector is reconstructed uniquely from*

$$pP \equiv A \pmod f, \qquad \deg p < N, \qquad u = p(T)e. \tag{10}$$

Proof. Necessity is (8). Conversely, P is a unit in $\mathbf{Q}[X]/(f)$, so (10) has a unique solution. With the resulting u, reduce (9) modulo f and use Lemma 2.2; this gives $tC_u = f - f_*$ as an equality because both sides have degree $< N$. Hence the full output is f_*. Substituting in $PC_u - A_u^2 = fD_u$ and in (9) gives $tD_u = P - P_*$, so the compressed output is P_*. □

3.1. **Two pins.** Assume that

$$f(0) = 0, \quad P(0) \neq 0, \qquad P(2) = 0, \quad f(2) \neq 0, \tag{11}$$

and that the displayed roots are simple. If a nonzero rank-one move retains the output roots $f_t(0) = 0$ and $P_t(2) = 0$, then $C_u(0) = 0$, hence $p(0) = A_u(0) = 0$. Evaluating the cofactor identity at 2 and using $D_u(2) = 0$ gives $A_u(2) = 0$. Therefore

$$X(X-2) \mid A_u, \qquad X^2(X-2)^2 \mid fP_t - f_tP. \tag{12}$$

When t may vary, the quotient in (12) must be a nonzero rational scalar times a square. By Theorem 3.1, this square-class filter is also sufficient for the fixed realization.

4. Arithmetic in a fixed pin direction

Let ϕ_0 span the zero eigenspace of T, and let ψ_2 span the 2-eigenspace of B. Suppose the two rational linear constraints below are independent and put

$$W = \{u : \langle u, \phi_0\rangle_H = 0,\ \langle u', \psi_2\rangle_{H'} = 0\}. \tag{13}$$

Then $\dim W = N - 2$. For $u \in W$ write

$$C_u = XE_u, \qquad D_u = (X-2)G_u, \qquad f = Xr, \quad P = (X-2)q. \tag{14}$$

Proposition 4.1 (Cyclic-coset integrality criterion)**.** *Assume $f = Xr$ with $r \in \mathbf{Z}[X]$, write $P = (X-2)q$ with $q \in \mathbf{Q}[X]$, and fix nonzero $u \in W$. Then*

$$f_t, P_t \in \mathbf{Z}[X] \iff tE_u \in \mathbf{Z}[X] \ \text{ and } \ q - tG_u \in \mathbf{Z}[X]. \tag{15}$$

For a nonzero rational polynomial $E = \sum_i e_i X^i$, define $\mathrm{cont}_{\mathbf{Q}}(E) > 0$ by

$$\sum_i \mathbf{Z}e_i = \mathrm{cont}_{\mathbf{Q}}(E)\mathbf{Z},$$

and set $\alpha_u = \mathrm{cont}_{\mathbf{Q}}(E_u)^{-1}$. Then

$$(15) \text{ is solvable with } t \neq 0 \iff [q] \in \langle [\alpha_u G_u] \rangle \text{ in } \mathbf{Q}[X]/\mathbf{Z}[X]. \tag{16}$$

Proof. The output formulas and (14) give $f_t = X(r - tE_u)$ and $P_t = (X-2)(q - tG_u)$, proving (15). By the definition of rational content, tE_u is integral exactly when $t = n\alpha_u$ for some $n \in \mathbf{Z}$. Substitution gives (16). Since the coefficients are rational, the generated subgroup is finite. Consequently, if a class is represented only by $n = 0$ in a first choice of representative, adding

the positive order of $[\alpha_u G_u]$ produces a nonzero representing integer and hence $t \neq 0$. □

Thus one direction is decided by Smith normal form or by a one-variable Chinese-remainder calculation. There is also a uniform denominator statement once directions are normalized.

Proposition 4.2 (Uniform content on a fixed pencil)**.** *Let $N \geq 3$, let $\Lambda = W \cap \mathbf{Z}^N$, and choose an integral basis S of Λ. There are positive integers L, Δ, depending only on the fixed pencil and this basis, such that for every primitive $z \in \mathbf{Z}^{N-2}$,*

$$\operatorname{cont}_{\mathbf{Z}}(LE_{Sz}) \mid \Delta. \tag{17}$$

Consequently the denominators of scalars producing integral full output are uniformly bounded on primitive integral directions.

Proof. Choose L so that the coefficients of LE_{Sz} are integral homogeneous quadrics $Q_1(z), \ldots, Q_m(z)$. They have no common projective zero over $\overline{\mathbf{Q}}$: otherwise $C_{Sz} = 0$, and the simple-spectrum adjugate formula

$$C_{Sz}(\theta) = f'(\theta) \frac{\langle Sz, \varphi_\theta \rangle_H^2}{\langle \varphi_\theta, \varphi_\theta \rangle_H}$$

would make Sz orthogonal to every eigenvector, hence zero. The projective Nullstellensatz supplies an integral Bézout certificate. Evaluating it at a primitive z shows that the gcd of all $Q_i(z)$ divides a fixed integer Δ, which is (17). The denominator assertion follows from Proposition 4.1. □

The primitive normalization is essential: replacing u by mu and t by t/m^2 represents the same matrix update. The proof also isolates a local obstruction. If at a prime p the coefficient quadrics of E have no projective base point modulo p, then some coefficient of E_{Sz} is a p-unit for every primitive z. Hence tE_{Sz} integral forces $v_p(t) \geq 0$; if G is integral at p, such a scalar cannot cancel a nonintegral coefficient of q. We call this a denominator-prime base-locus obstruction, not a ramification theorem.

5. An infinite quartic rank-one obstruction

For $K \geq 3$ set

$$r_K = X^2 - (K-1)X - 1, \quad f_K = X r_K, \quad P_K = (X-2)\left(X + \frac{1}{K}\right). \tag{18}$$

The reciprocal lift is

$$B_K(z) = z^2 r_K(z + z^{-1}) = z^4 - (K-1)z^3 + z^2 - (K-1)z + 1. \tag{19}$$

This is a genuine quartic Salem polynomial. Indeed r_K has roots $y_K > 2$ and $\alpha_K \in (-2, 0)$ with $y_K \alpha_K = -1$ and $K - 1 < y_K < K$; its discriminant lies strictly between consecutive squares. The signs of $y_K^2 - 4$ at the two real embeddings then show that the reciprocal lift is irreducible. The same conclusion follows directly from the classical quartic criterion [16].

Definition 5.1 (Normal height)**.** If a Salem polynomial of degree $2s$ is written $R(z) = z^s r(z + z^{-1})$ and $f_R = Xr$, a strict integral normal interlacer is a polynomial $b \in \mathbf{Z}[X]$ of degree $s-1$, with negative leading coefficient and one simple zero in every gap between the compact roots of f_R. Its normal height is

$$\kappa(R) = \min_b(-\operatorname{lc} b).$$

For the quartic B_K, this says precisely that $b = -\kappa X + c$ is primitive and its zero lies strictly between α_K and 0. The corresponding monic compressed polynomial is

$$P_b = -\frac{(X-2)b}{\kappa}. \tag{20}$$

Proposition 5.2 (Exact height)**.** *For every $K \geq 3$, $\kappa(B_K) = K$.*

Proof. The zero c/κ lies in $(\alpha_K, 0) = (-1/y_K, 0)$, so $c < 0$ and $|c|/\kappa < 1/y_K$. Primitivity gives $|c| \geq 1$, while $y_K > K - 1$; hence $\kappa \geq K$. The polynomial $-KX - 1$ has zero $-1/K$ in the required interval and attains the bound. □

Theorem 5.3 (Quartic rank-one obstruction)**.** *For every $K \geq 5$, no nonzero signed rational rank-one update of the fixed pair (18) can simultaneously*

(i) *preserve the output roots 0 and 2,*
(ii) *make both output polynomials integral, and*
(iii) *have the trace-root signature of a genuine quartic Salem polynomial.*

The threshold is sharp: $K = 4$ admits such an output.

Proof. Write an integral pinned target as

$$f_* = Xr_*, \quad r_* = X^2 + aX + b, \qquad P_* = (X-2)(X+n), \qquad a, b, n \in \mathbf{Z}.$$

Theorem 3.1 and (12) reduce the cross identity to

$$r_K(X+n) - r_*\left(X + \frac{1}{K}\right) = \mu X(X-2) \tag{21}$$

for some $\mu \in \mathbf{Q}^\times$. Coefficient comparison gives

$$b = -Kn, \qquad a = \frac{3Kn - 2K^2 + K - 2}{2K+1}, \qquad \mu = -\frac{(K-1)(Kn-1)}{K(2K+1)}. \tag{22}$$

Since $2K+1$ is odd, $a \in \mathbf{Z}$ is equivalent to

$$2K + 1 \mid 3(n+2). \tag{23}$$

Put $g = \gcd(3, 2K+1)$ and $L = (2K+1)/g$. Thus $n = -2 + L\ell$.

If $\ell = 0$, then $r_* = (X-2)(X-K)$, so the root 2 collides. If $\ell \leq -1$, evaluation of (21) at 2 gives

$$r_*(2) = \frac{(5-2K)(n+2)}{2 + 1/K} > 0,$$

whereas a monic quadratic having one root in $(-2,2)$ and one root above 2 must be negative at 2. Finally, if $\ell \geq 1$, then $n \geq 3$ for $K \geq 5$, and direct substitution gives

$$r_*(-2) = \frac{4K^2 + 6K + 8 - Kn(2K+7)}{2K+1} < 0.$$

The desired two roots both lie to the right of -2, which would require $r_*(-2) > 0$. All cases are excluded.

For $K = 4$, take

$$n = 1, \qquad r_* = X^2 - 2X - 4, \qquad \mu = -\frac{1}{4}.$$

Its roots are $1 \pm \sqrt{5}$, one in $(-2,2)$ and the other above 2. Moreover r_* is irreducible because its discriminant is 20, and $r_*(2)r_*(-2) = -16 < 0$ shows that $y^2 - 4$ is not a square in its trace field. Hence the reciprocal lift is an irreducible quartic Salem polynomial. Equation (21) and Theorem 3.1 supply the actual signed rational update. □

The obstruction is not a failure of irreducibility or of a thin-set argument. The integral lattice itself misses the open Salem chamber. It therefore cannot be removed by strong approximation or Hilbert irreducibility.

6. Rank-two Gram–Plücker surgery

Let $U = (u_1, u_2) \in M_{N,2}(\mathbf{Q})$, let U' be obtained by deleting its first row, and let $S = S^{\mathsf{T}} \in M_2(\mathbf{Q})$. Define

$$A = e^{\mathsf{T}} \operatorname{adj}(XI - T)U \in M_{1,2}(\mathbf{Q}[X]), \tag{24}$$

$$C = U^{\mathsf{T}} H \operatorname{adj}(XI - T)U \in \operatorname{Sym}_2(\mathbf{Q}[X]), \tag{25}$$

$$D = (U')^{\mathsf{T}} H' \operatorname{adj}(XI - B)U' \in \operatorname{Sym}_2(\mathbf{Q}[X]). \tag{26}$$

Polarization of Lemma 2.2 gives

$$\boxed{PC - A^{\mathsf{T}}A = fD}. \tag{27}$$

At each simple zero of f, the matrix C has rank at most one, hence $f \mid \det C$; similarly $P \mid \det D$. Put

$$\Phi_C = \frac{\det C}{f}, \qquad \Phi_D = \frac{\det D}{P}. \tag{28}$$

Taking determinants in (27) yields the polynomial

$$\boxed{\Psi_U = \frac{A \operatorname{adj}(C) A^{\mathsf{T}}}{f} = P\Phi_C - f\Phi_D \in \mathbf{Q}[X]}. \tag{29}$$

For the update

$$T_S = T + USU^{\mathsf{T}}H, \tag{30}$$

the 2×2 determinant lemma, valid also for singular S, gives

$$f_S = f - \operatorname{tr}(SC) + \det(S)\Phi_C, \qquad P_S = P - \operatorname{tr}(SD) + \det(S)\Phi_D. \tag{31}$$

Theorem 6.1 (Gram–Plücker law)**.** *The cross polynomial of* (30) *is*

$$\boxed{fP_S - f_S P = ASA^{\mathsf{T}} - \det(S)\Psi_U.} \tag{32}$$

The product $\det(S)\Psi_U$ *is invariant under changing the basis of the update plane.*

Proof. Substitute (31), use $PC - fD = A^{\mathsf{T}}A$, and use $P\Phi_C - f\Phi_D = \Psi_U$. Under $U \mapsto UL$ and $S \mapsto L^{-1}SL^{-\mathsf{T}}$, Ψ_U is multiplied by $\det(L)^2$ and $\det S$ by $\det(L)^{-2}$. □

When $S = tcc^{\mathsf{T}}$ has rank one, the determinant term vanishes and (32) is the square law. For nonsingular S, rational congruence diagonalization expresses the first term as a binary Gram form, but the nonzero exterior correction remains. This is why the rank-two image is not described by a bare binary norm form.

Theorem 6.2 (Exact rank-at-most-two assignment)**.** *Let* f_*, P_* *be monic rational polynomials of degrees* $N, N-1$*. The following are equivalent.*

(a) *The pair* (f_*, P_*) *is the full–minor output of* $T + USU^{\mathsf{T}}H$ *for rational* $U \in M_{N,2}(\mathbf{Q})$ *and symmetric rational* S*.*

(b) *There are* $A_1, A_2 \in \mathbf{Q}[X]_{<N}$ *and* $S = S^{\mathsf{T}} \in M_2(\mathbf{Q})$ *such that, after uniquely reconstructing*

$$p_i P \equiv A_i \pmod f, \qquad C_{ij} = \operatorname{rem}_f(p_i p_j P), \qquad D = \frac{PC - A^{\mathsf{T}}A}{f},$$

and then setting

$$\Phi_C = \frac{\det C}{f}, \qquad \Phi_D = \frac{\det D}{P}, \qquad \Psi = P\Phi_C - f\Phi_D,$$

one has

$$fP_* - f_*P = ASA^{\mathsf{T}} - \det(S)\Psi. \tag{33}$$

The update has rank exactly two when p_1, p_2 *are independent modulo* f *and* $\det S \neq 0$*.*

Proof. Necessity is Theorem 6.1. Conversely set $u_i = p_i(T)e$ and let $(\widetilde{f}, \widetilde{P})$ be the actual output. Equations (32) and (33) imply

$$f(\widetilde{P} - P_*) = P(\widetilde{f} - f_*).$$

Since $(f, P) = 1$, the polynomial f divides $\widetilde{f} - f_*$. The latter has degree at most $N-1$, so it vanishes. The compressed difference then vanishes as well. The final assertion follows because H is nonsingular, the columns of U are independent, and S is nonsingular. □

6.1. **Fixed-plane feasibility as one quadric.** The converse above also turns fixed-plane feasibility into an exact local–global calculation rather than a search through update matrices.

Proposition 6.3 (Fixed-plane linear–quadric reduction)**.** *Fix a rational two-plane U and write*

$$S = \begin{pmatrix} r & s \\ s & u \end{pmatrix}, \qquad d = \det S.$$

For prescribed rational monic polynomials f_*, P_*, coefficient matching in (31) is a rational affine linear system

$$\mathcal{L}_{U,f_*,P_*}(r, s, u, d) = b. \tag{34}$$

The pair (f_*, P_*) is attainable in this plane if and only if (34) has a rational solution satisfying

$$d = ru - s^2. \tag{35}$$

After rationally parametrizing the affine solution space of (34), feasibility is therefore the existence of a rational point on one explicit affine quadric.

If the projective closure of the residual quadric is nonsingular and has positive dimension, feasibility is equivalent to feasibility over $\mathbf{R}$ and over every $\mathbf{Q}_p$. Singular and zero-dimensional residual problems reduce, by rational linear algebra, to a lower-rank quadratic, a discriminant test, or a linear equation.

Proof. The first assertion is immediate from

$$f_S = f - \operatorname{tr}(SC) + d\Phi_C, \qquad P_S = P - \operatorname{tr}(SD) + d\Phi_D :$$

all coefficients are affine linear in (r, s, u, d), and the only condition lost by treating d as an independent variable is (35). Conversely, a common solution defines a symmetric S with determinant d, and (31) gives the prescribed pair.

Gaussian elimination parametrizes every nonempty rational affine solution space. Substitution in (35) leaves one quadratic equation. For a nonsingular projective closure of positive dimension, the Hasse–Minkowski theorem gives a rational projective point exactly when there is a point over every completion; weak approximation for quadrics then moves a rational point away from the hyperplane at infinity. The degenerate cases are obtained by splitting off the radical and solving the remaining quadratic or linear equation; see [5]. □

7. Output pins versus fixed eigenspaces

If only $f_S(0) = 0$ and $P_S(2) = 0$ are required, then tautologically

$$X(X - 2) \mid fP_S - f_S P. \tag{36}$$

For indefinite S, cancellation can preserve an eigenvalue while changing its eigendirection. Thus (36) need not contain squared pin factors.

Pointwise eigenspace preservation is different. The following theorem is stated over a general field because it is linear algebra rather than positivity.

Lemma 7.1 (Active congruence factorization)**.** *Let the ground field have characteristic different from* 2*, let* $H = H^{\mathsf{T}}$ *be nonsingular, and suppose* $H\Delta = \Delta^{\mathsf{T}}H$*. If* $r = \operatorname{rank}\Delta$*, then*

$$\Delta = USU^{\mathsf{T}}H$$

for a full-column-rank $N \times r$ *matrix* U *and a nonsingular symmetric* $r \times r$ *matrix* S*.*

Proof. The matrix $M = \Delta H^{-1}$ is symmetric of rank r. A symmetric matrix over a field of characteristic different from 2 has a rank factorization $M = USU^{\mathsf{T}}$ obtained by restricting its bilinear form to a nondegenerate complement of its radical. Here U has full column rank and S is symmetric and nonsingular. Multiplication by H gives the claim. □

Theorem 7.2 (All-rank fixed-eigenspace barrier)**.** *Let the ground field have characteristic different from* 2*. Suppose* $H = H^{\mathsf{T}} = \operatorname{diag}(h, H')$ *is nonsingular and* $HT = T^{\mathsf{T}}H$*. Let*

$$T\phi_0 = \lambda_0\phi_0, \qquad B\psi = \lambda_2\psi, \qquad e^{\mathsf{T}}H\phi_0 \neq 0.$$

Let $\Delta = USU^{\mathsf{T}}H$ *have effective rank* r*, with the inactive null directions removed so that* U *has full column rank and* S *is nonsingular. If* $T + \Delta$ *fixes* ϕ_0 *with eigenvalue* λ_0 *and its first compression fixes* ψ *with eigenvalue* λ_2*, then*

$$r \leq N - 2. \tag{37}$$

More generally, if a d*-dimensional eigenspace of* B *is fixed pointwise, then* $r \leq N - 1 - d$*.*

Proof. The full pin gives $0 = USU^{\mathsf{T}}H\phi_0$, hence $U^{\mathsf{T}}H\phi_0 = 0$ by injectivity of U and S. Deletion of the first coordinate is injective on $\operatorname{im} U$. Indeed, if $Uc = ae$, then

$$0 = (Uc)^{\mathsf{T}}H\phi_0 = ae^{\mathsf{T}}H\phi_0,$$

so $a = 0$ and then $c = 0$. Thus the deleted matrix U' still has rank r. The compressed pin gives $(U')^{\mathsf{T}}H'\psi = 0$, placing $\operatorname{im} U'$ in one hyperplane of an $(N-1)$-dimensional space. This proves (37). For a pointwise fixed d-space, the image lies in its H'-orthogonal complement, of dimension $N - 1 - d$. □

Corollary 7.3 (Order-three collapse)**.** *For* $N = 3$*, every finite-rank* H*-self-adjoint perturbation fixing the full eigenline and the compressed eigenline as above has effective rank at most one. Hence Theorem 5.3 survives every nominal-rank attempt under pointwise fixed-eigenspace pins.*

Proof. Apply Lemma 7.1 and Theorem 7.2 with $d = 1$. □

There is one important situation in which output pins automatically become eigenspace pins. For a positive form H, call an H-self-adjoint perturbation Δ H-positive (respectively H-negative) semidefinite when the symmetric matrix $H\Delta$ is positive (respectively negative) semidefinite.

Lemma 7.4 (Semidefinite equality forces an eigenline pin)**.** *Let $A = A^{\mathsf{T}}$ and $E = E^{\mathsf{T}}$ be real matrices with $E \succeq 0$ or $E \preceq 0$. Suppose that the ordered eigenvalue $\lambda_j(A)$ is simple and*

$$\lambda_j(A + E) = \lambda_j(A).$$

Then E annihilates an eigenvector of A belonging to $\lambda_j(A)$.

Proof. Assume first that $E \succeq 0$. The ordered eigenvalue $\lambda_j(A + tE)$ is nondecreasing for $0 \le t \le 1$; see, for example, [3]. Equality at the endpoints makes it constant. If v is a unit eigenvector for the simple initial eigenvalue, the Hellmann–Feynman formula gives

$$\frac{d}{dt}\lambda_j(A + tE)\Big|_{t=0} = v^{\mathsf{T}} E v = 0.$$

Positive semidefiniteness implies $Ev = 0$. The negative-semidefinite case follows by applying the same argument to $-E$. □

Theorem 7.5 (All-rank semidefinite obstruction in the strict slice)**.** *Let $K \ge 5$ and let Δ be a rational H-self-adjoint perturbation of a positive rational realization of the pair (18). Suppose that $H\Delta$ is semidefinite and that its output pair is*

$$\begin{gathered} f_* = X(X^2 - aX - K), \qquad P_* = (X - 2)(X + 1), \\ a \in \mathbf{Z}, \qquad \frac{K - 4}{2} < a < K - 1. \end{gathered} \tag{38}$$

Then no such perturbation exists.

Proof. Conjugate by $H^{1/2}$. At both endpoints, 0 is the simple middle full eigenvalue: the other two full roots consist of one point in $(-2, -1)$ and one point above 2. Likewise, 2 is the simple maximal eigenvalue of both compressed matrices. Lemma 7.4, applied to the full matrix and to its principal compression, shows that Δ fixes the initial full zero eigenline and the initial compressed 2-eigenline pointwise. Theorem 7.2 now gives $\operatorname{rank} \Delta \le 1$. The zero update cannot change $P_K = (X - 2)(X + 1/K)$ to P_*, while every nonzero rank-one update is excluded by Theorem 5.3. □

Remark 7.6. The threshold is sharp. The $K = 4$ exception in the proof of Theorem 5.3 has square-law scalar $-1/4$ and hence is a negative-semidefinite rational rank-one update. The theorem above is not a claim about arbitrary semidefinite perturbations: the integral output pair, the two ordered pins, and the strict Salem signature in (38) are essential.

In the positive rational setting of (11), take an active rank-two representation. Pointwise preservation is columnwise orthogonality, so the spectral adjugate formula gives $A_i(0) = 0$. At the compressed pin one has $D(2) = 0$; evaluating (27) at 2 gives $A(2)^{\mathsf{T}} A(2) = 0$, hence $A_i(2) = 0$ over $\mathbf{Q}$. Thus every A_i is divisible by $\eta = X(X - 2)$. Since $A_i(0) = p_i(0)P(0)$ and $P(0) \neq 0$,

the full pin also gives $p_i(0) = 0$; hence every entry of C is divisible by X and, for a 2×2 matrix, so is every entry of $\operatorname{adj}(C)$. Writing $f = Xr$ gives

$$\Psi = \frac{A \operatorname{adj}(C) A^{\mathsf{T}}}{f} = \eta^2 \frac{N(X)}{r(X)}$$

for some $N \in \mathbf{Q}[X]$. Since Ψ is a polynomial and $\gcd(r, \eta) = 1$, one has $r \mid N$, and therefore $\eta^2 \mid \Psi$. Thus both terms in (32) carry the squared pin divisor. The constructions below deliberately require only output pins; they fix the zero eigenline but change the compressed 2-eigenline.

8. A rank-free obstruction with the canonical form

We now fix more structure. Put

$$\mathcal{B}_K = 2K^3 - 6K^2 + 3K - 1, \qquad b_1 = \frac{\mathcal{B}_K}{K^2}, \qquad b_2 = \frac{2K(K-1)(2K-5)}{\mathcal{B}_K^2}, \tag{39}$$

and let T_K be the rational Jacobi realization of (f_K, P_K) displayed in Appendix B. Its positive symmetrizing form is

$$H_K = \operatorname{diag}(1, b_1^{-1}, (b_1 b_2)^{-1}). \tag{40}$$

The next obstruction is independent of an update plane and of the effective rank, but it is tied to this fixed form and to the first-coordinate compression.

Theorem 8.1 (Rank-free same-form norm obstruction)**.** *Let $a, m \in \mathbf{Q}$. Suppose that a rational H_K-self-adjoint matrix T_* has characteristic pair*

$$f_* = X(X^2 - aX - m), \qquad P_* = (X-2)(X+1). \tag{41}$$

Then the affine conic

$$Y^2 + 2K(K-1)(2K-5)Z^2 = K^2 \mathcal{B}_K (a + m - 3) \tag{42}$$

has a rational point. Consequently, (42) is necessary for a rational H_K-self-adjoint perturbation of T_K of any effective rank.

Proof. Every rational H_K-self-adjoint matrix has the form

$$T_* = \begin{pmatrix} x & y & z \\ b_1 y & u & v \\ b_1 b_2 z & b_2 v & w \end{pmatrix}.$$

The compressed characteristic polynomial in (41) gives

$$u + w = 1, \qquad uw - b_2 v^2 = -2,$$

while the full trace gives $x = a - 1$. Comparing the coefficient of X in $\det(XI - T_*) = X^3 - aX^2 - mX$ yields

$$b_1 y^2 + b_1 b_2 z^2 = a + m - 3.$$

Multiplication by $K^2 \mathcal{B}_K$ and the substitution $Y = \mathcal{B}_K y$, $Z = z$ give (42).

The conclusion does not depend on a presentation of the perturbation. The difference of two rational H_K-self-adjoint matrices is again H_K-self-adjoint,

and Lemma 7.1 represents it as $USU^{\mathsf{T}}H_K$ with effective rank equal to its matrix rank. □

Corollary 8.2 (An infinite fixed-form, all-rank Salem obstruction)**.** *Let $p > 5$ be prime with*

$$p \equiv 5 \text{ or } 7 \pmod 8, \qquad K = \frac{p+5}{2},$$

and set $a = K - 4$, $m = K$. The pair

$$f_* = X\big(X^2 - (K-4)X - K\big), \qquad P_* = (X-2)(X+1) \tag{43}$$

is a genuine height-one Salem target, but it cannot be reached from the fixed canonical realization (T_K, H_K) by a rational H_K-self-adjoint perturbation of any rank.

Proof. Here $2K - 5 = p$, $a + m - 3 = p - 2$, and $\mathcal{B}_K \equiv 1/4 \pmod p$. The right side of (42) is therefore a p-adic unit whose residue square class is

$$K^2\mathcal{B}_K(p-2) \equiv -\frac{25}{8} = -2\left(\frac{5}{4}\right)^2 \pmod p.$$

Since $(-2/p) = -1$ for $p \equiv 5, 7 \pmod 8$, the conic has no $\mathbf{Q}_p$-point. Indeed, in an equation $Y^2 + pcZ^2 = d$ with $c, d \in \mathbf{Z}_p^\times$, the two summands have valuations of opposite parity. A negative valuation of either variable would give the left side negative valuation, so $Y, Z \in \mathbf{Z}_p$; reduction modulo p would then make the nonsquare d a square.

It remains to check the target. For $q = X^2 - (K-4)X - K$ one has

$$q(-2) = K - 4 > 0, \qquad q(-1) = -3 < 0, \qquad q(2) = -3(K-4) < 0.$$

Thus the roots lie in $(-2, -1)$ and $(2, \infty)$. Its discriminant is $(K-2)^2 + 12$, which is not a square for $K \geq 6$: a square would give $(s-K+2)(s+K-2) = 12$, whose only positive same-parity factorization leads to $K = 4$. Moreover

$$q(2)q(-2) = -3(K-4)^2 < 0,$$

so $y^2 - 4$ is not a square in the quadratic trace field. Hence the reciprocal lift is irreducible and Salem. The primitive normal interlacer $-(X+1)$ gives height one. The obstruction now follows from Theorem 8.1. □

The restriction to the canonical form is essential. Strict interlacing provides other rational symmetrizable realizations of the target pair by Lemma 2.1; the corollary prohibits reaching it while keeping the particular form H_K and the particular first compression fixed. By Dirichlet's theorem, the two residue classes used above contain infinitely many primes.

9. A complete height-one target slice

Fix $K \geq 5$, set $T = T_K$, and use the canonical update plane

$$U = (Te, T^2e), \qquad p_1 = X, \quad p_2 = X^2. \tag{44}$$

Consider the target slice

$$f_{a,m} = X(X^2 - aX - m), \qquad P_* = (X-2)(X+1), \qquad a, m \in \mathbf{Q}. \tag{45}$$

For this plane direct cofactor calculation gives

$$\Phi_C = \frac{(K-1)(2K-5)}{K^2} X, \tag{46}$$

$$\Phi_D = \frac{2(K-1)(2K-5)}{K^3}, \tag{47}$$

$$\Psi = \frac{X^2(K-1)(2K-5)((K-2)X-1)}{K^3}. \tag{48}$$

Theorem 9.1 (Exact rational image of the slice)**.** *For $K \geq 5$, a rational symmetric rank-at-most-two update in the plane (44) has output (45) if and only if*

$$m = K, \qquad K(K-1)(2K-5)(K-1-a)(2a+K-4) \in \mathbf{Q}^2, \tag{49}$$

where $\mathbf{Q}^2 = \{q^2 : q \in \mathbf{Q}\}$ includes zero. For $K > 4$, the target has strict Salem interlacing precisely when

$$\frac{K-4}{2} < a < K-1. \tag{50}$$

Proof. Write $S = \left(\begin{smallmatrix} r & s \\ s & u \end{smallmatrix}\right)$ and $d = ru - s^2$. Substitute (46)–(48) and the cofactor matrices from (25)–(26) into (31). Lemma A.1 records the coefficient solution explicitly. In particular, coefficient comparison forces

$$m = K. \tag{51}$$

The remaining determinant condition is a genuine quadratic in d whose discriminant is

$$-16K^5(K-1)^3(2K-5)^3(a-K+1)(2a+K-4). \tag{52}$$

Removing the evident square factor $16K^4(K-1)^2(2K-5)^2$ gives exactly the square class in (49). The coefficient of d^2 is $-(K-2)^2(K-1)^2(2K-5)^2 \neq 0$. Conversely, if this class is a square (including zero), the quadratic has a rational root d, back-substitution gives a rational symmetric S, and Theorem 6.2 supplies the desired update. This proves the classification.

For $m = K$, put $q(X) = X^2 - aX - K$. Since $q(0) < 0$, its negative root lies in $(-2, -1)$ precisely when $q(-2) > 0$ and $q(-1) < 0$, namely

$$2a + 4 - K > 0, \qquad 1 + a - K < 0.$$

These are exactly the two inequalities in (50). They also imply $q(2) < 0$ for $K > 4$, so the positive root lies above 2. Thus the compressed roots $-1, 2$ strictly interlace the three full roots. □

The eliminant computation in the proof is elementary but long. The source package includes a symbolic certificate that derives (51) and (52) from the cofactor formulas without floating-point arithmetic.

Proposition 9.2 (Parametrization, finite square sieve, and forced signature)**.** *Fix an integer $K \geq 5$, put*

$$\Gamma_K = K(K-1)(2K-5), \qquad n = K-1-a,$$

and restrict to $m = K$. The rational image condition of Theorem 9.1 is

$$Y^2 = \Gamma_K n(3(K-2)-2n). \tag{53}$$

Apart from $(n, Y) = (0, 0)$, all rational points of (53) are parametrized by $\ell \in \mathbf{Q}$ as

$$n = \frac{3\Gamma_K(K-2)}{\ell^2 + 2\Gamma_K}, \qquad a = K-1-n, \qquad Y = \ell n. \tag{54}$$

If $a \in \mathbf{Z}$, strict Salem interlacing is equivalent to

$$1 \leq n \leq \left\lfloor \frac{K+1}{2} \right\rfloor. \tag{55}$$

Consequently, the strict integral image at fixed K is obtained by the finite test of retaining exactly those n in (55) for which the right side of (53) is an integer square.

Every rational realization in the strict chamber with nonzero determinant has $\det S < 0$. For a strict integral target every realizing branch has nonzero determinant and is therefore a genuine indefinite rank-two update.

Proof. After substituting $a = K-1-n$, the square class in (49) becomes the right side of (53). The line $Y = \ell n$ through $(0, 0)$ gives (54), and every point with $n \neq 0$ has $\ell = Y/n$. The inequalities (50) become $0 < n < (K+2)/2$, which for integral n is precisely (55). An integer which is a square in $\mathbf{Q}$ is an integer square.

In the normalization of Lemma A.1, the determinant eliminant after $a = K-1-n$ may be chosen as

$$\begin{aligned} \mathcal{Q}_{K,n}(d) = & -(K-2)^2(K-1)^2(2K-5)^2d^2 \\ & -2K^2(K-1)(2K-5) \\ & \quad \times \Big((2K^2-7K+2)n + 3(K-1)(K-2)\Big)d \\ & -K^4\big((2K+1)n - 3K + 3\big)^2. \end{aligned} \tag{56}$$

For $K \geq 5$ and $n > 0$, the coefficient in parentheses on the second line is positive. Hence the sum of the two roots is negative and their product is nonnegative. When the square condition holds the roots are rational, so every nonzero root is negative. In the integral strict range the constant term cannot vanish: $(2K+1)n = 3(K-1)$ would force $K = 4$ if $n = 1$, while for $n \geq 2$ the left side exceeds the right side. Both determinant branches are therefore strictly negative. □

Corollary 9.3 (Local–global square test on the exact slice)**.** *Put*

$$D(K, a) = K(K-1)(2K-5)(K-1-a)(2a+K-4).$$

The canonical fixed-plane surgery exists over $\mathbf{Q}$ *if and only if* $D(K,a) \in \mathbf{Q}^2$. *Equivalently,*

$$D(K,a) \geq 0 \quad \textit{and} \quad D(K,a) \in \mathbf{Q}_p^2 \quad \textit{for every prime } p.$$

Only the real place and the primes dividing the numerator or denominator of $2D(K,a)$ *need be tested.*

Proof. The first statement is Theorem 9.1. A rational number is a rational square exactly when it is nonnegative at the real place and a square in every $\mathbf{Q}_p$. For the finite assertion, write $D = A/B$ in lowest terms. If the indicated local tests pass, then every prime valuation of A and B is even and the real sign is positive; hence both A and B are squares and $D \in \mathbf{Q}^2$. The converse is immediate. This is the elementary one-dimensional case of the usual local square criterion; compare [5]. □

Example 9.4 (One-prime all-depth solvability does not descend)**.** For $K = 6$, $a = 2$, the target is

$$f_* = X(X^2 - 2X - 6), \qquad P_* = (X-2)(X+1).$$

The same-form norm equation is

$$(233y)^2 + 420z^2 = 41940,$$

and has no point over $\mathbf{Q}_7$; hence no rational H_6-self-adjoint surgery of any rank produces this pair. In the canonical rank-two plane, however, coefficient elimination gives

$$r = -\frac{3(1925d + 11036)}{70}, \quad s = \frac{485d + 2808}{30}, \quad u = -\frac{665d + 3888}{210},$$

and

$$F(d) = 1225d^2 + 24570d + 46656 = 0. \tag{57}$$

Modulo 13, a unit multiple of F is $4d^2 + 3$. The root $d = 3$ is simple because $8d \equiv 11 \pmod{13}$. Hensel's lemma supplies a compatible $d \in \mathbf{Z}_{13}$ and hence a canonical-plane solution over $\mathbf{Z}_{13}$, reducing at the first level to

$$S \equiv \begin{pmatrix} 4 & 3 \\ 3 & 3 \end{pmatrix} \pmod{13}.$$

Thus solutions exist modulo every 13^r, although no rational solution exists. Consistently,

$$\operatorname{disc} F = 315^2 \cdot 3780, \qquad D(6,2) = 3780 \notin \mathbf{Q}^2.$$

This example separates one-prime infinite-depth lifting from a genuine global local–global test.

10. The canonical near-shadow obstruction

We next ask whether the exact slice can preserve the exterior trace up to a vanishing error. The answer is negative, for an elliptic reason.

Theorem 10.1 (Arithmetic exclusion of the canonical near-shadow)**.** *For an integer $K \geq 5$, the formally closest integral member of the target slice is obtained by taking $a = K-2$. It is a genuine height-one quartic Salem target, but it is not in the rational image of the canonical plane. More precisely, its square condition is*

$$Y^2 = K(K-1)(2K-5)(3K-8), \tag{58}$$

whose affine rational solutions are exactly

$$(0,0), (1,0), \left(\frac{5}{2}, 0\right), \left(\frac{8}{3}, 0\right), (2, \pm 2), (4, \pm 12).$$

In particular, none has $K \geq 5$.

Proof. Substitution of $a = K-2$ in (49) gives (58). Put

$$z = 24 - \frac{40}{K}, \qquad w = \frac{40Y}{K^2}. \tag{59}$$

Away from $K = 0$ this is birational, with inverse

$$K = \frac{40}{24-z}, \qquad Y = \frac{40w}{(24-z)^2},$$

and transforms (58) into

$$E: \quad w^2 = (z+16)(z-8)(z-9). \tag{60}$$

The point $(K, Y) = (0, 0)$ extends to the point at infinity of E.

For completeness, we record a short 2-isogeny descent. Set $x = z - 9$. Then

$$E_1: \quad w^2 = x(x+1)(x+25) = x^3 + 26x^2 + 25x,$$

and the curve 2-isogenous through $(0, 0)$ is

$$E_1': \quad v^2 = u^3 - 52u^2 + 576u = u(u-16)(u-36).$$

For the standard descent maps α, α', one has

$$2^{\operatorname{rank} E_1(\mathbf{Q})} = \frac{|\alpha(E_1(\mathbf{Q}))|\,|\alpha'(E_1'(\mathbf{Q}))|}{4}; \tag{61}$$

see [6, Chapter III]. The first image is contained in the four square classes $\{1, -1, 5, -5\}$, all attained at O, $(-1, 0)$, $(5, 30)$, and $(-5, 20)$.

We claim that $\alpha'(E_1'(\mathbf{Q})) = \{1\}$. Since $u(u-16)(u-36) < 0$ for $u < 0$, only the positive square classes $1, 2, 3, 6$ can occur. If $c \in \{2, 3, 6\}$ occurred, the homogeneous-space criterion would give coprime integers M, e and an integer N satisfying

$$N^2 = cM^4 - 52M^2e^2 + \frac{576}{c}e^4. \tag{62}$$

For $c = 2$, reduction modulo 4 forces $M = 2m$ and e odd, and then $N = 4n$ with

$$n^2 = 2m^4 - 13m^2e^2 + 18e^4.$$

Modulo 8 the right side is 7 if m is odd and 2 or 6 if m is even, never a square. For $c = 3$ or 6, reduction modulo 3 forces $3 \mid Me$. Coprimality leaves exactly one of M, e divisible by 3; modulo 9, (62) then gives respectively $N^2 \equiv 3$ or 6 (mod 9), again impossible. Thus $|\alpha'| = 1$, and (61) gives $\operatorname{rank} E_1(\mathbf{Q}) = 0$.

The eight visible rational points are

$$O,\ (-25, 0),\ (-1, 0),\ (0, 0),\ (-5, \pm 20),\ (5, \pm 30).$$

Both 7 and 11 are primes of good reduction, and direct counting gives $\#E_1(\mathbf{F}_7) = \#E_1(\mathbf{F}_{11}) = 8$. For every prime ℓ, the ℓ-primary rational torsion injects at one of these two reductions; hence $\#E_1(\mathbf{Q})_{\mathrm{tors}}$ divides $\gcd(8, 8) = 8$. The displayed eight points are therefore all of $E_1(\mathbf{Q})$. This is the curve of Cremona label $240a3$ (modern LMFDB label $240.c2$); the same rank and torsion determination appears in [12, Proposition 5]. Applying the inverse birational map gives the asserted solutions of (58).

Finally, for $a = K - 2$, the trace polynomial $q_K = X^2 - (K-2)X - K$ has one root in $(-2, -1)$ and one above 2. Its discriminant $K^2 + 4$ lies strictly between K^2 and $(K+1)^2$, while $q_K(2)q_K(-2) = K(8 - 3K) < 0$. Thus its reciprocal lift is irreducible and Salem, and the primitive interlacer $-(X+1)$ has height one. □

Corollary 10.2 (No vanishing exterior shift in the integral slice)**.** *Let*

$$y_K = \frac{K - 1 + \sqrt{(K-1)^2 + 4}}{2}, \qquad y_{a,K} = \frac{a + \sqrt{a^2 + 4K}}{2}.$$

Along any sequence of strictly interlacing integral targets in the rational image of the canonical plane with $K \to \infty$,

$$\liminf |y_{a,K} - y_K| \geq \frac{1}{2}.$$

In particular, this complete height-one slice contains no $o(1)$ exterior shadow surgery.

Proof. For the unique nearest integer $a_0 = K - 2$,

$$y_{a_0,K} = \frac{K - 2 + \sqrt{K^2 + 4}}{2}, \qquad y_{a_0,K} - y_K = -K^{-2} - K^{-3} + 2K^{-4} + O(K^{-5}).$$

Theorem 10.1 excludes a_0 from the rational image for every $K \geq 5$. In the strict chamber $a > 0$ and

$$\frac{\partial y_{a,K}}{\partial a} = \frac{1}{2}\left(1 + \frac{a}{\sqrt{a^2 + 4K}}\right) > \frac{1}{2}.$$

Every other integral a satisfies $|a - a_0| \geq 1$, so the mean-value theorem and the triangle inequality give

$$|y_{a,K} - y_K| > \frac{1}{2} - |y_{a_0,K} - y_K| = \frac{1}{2} - o(1).$$

□

Corollary 10.3 (Fixed bounded shifts are finite)**.** *Fix an integer $h \le 0$ and put $a = K - 2 + h$. There are only finitely many integers $K \ge 5$ for which this target lies in the rational image of the canonical plane. Consequently, no infinite integral image family with bounded h exists.*

Proof. The square condition becomes

$$Y^2 = (1-h)K(K-1)(2K-5)(3K+2h-8).$$

For fixed $h \le 0$ the four roots on the right are distinct, so this is a nonsingular affine genus-one curve. Siegel's theorem gives only finitely many integral points; see [15, Chapter IX]. A bounded set of integer shifts is a finite union of such cases. □

11. Exact Salem height collapse in rank two

We first record a small sporadic certificate, then give an infinite family.

Proposition 11.1 (The $K = 5$ certificate)**.** *For $K = 5$ and the plane (44), take*

$$S = \begin{pmatrix} -159/2 & 21 \\ 21 & -11/2 \end{pmatrix}, \qquad \det S = -\frac{15}{4}.$$

Then

$$f_S = X(X^2 - X - 5), \qquad P_S = (X-2)(X+1). \tag{63}$$

The reciprocal lift

$$z^4 - z^3 - 3z^2 - z + 1 \tag{64}$$

is an irreducible Salem polynomial, the update has rank exactly two, and the output normal height is one. This output is impossible in rank one.

Proof. Substitution in (31) gives (63). The trace roots are $(1 \pm \sqrt{21})/2$, so

$$-2 < \frac{1-\sqrt{21}}{2} < -1 < 0 < 2 < \frac{1+\sqrt{21}}{2}.$$

Irreducibility follows either from the quadratic trace polynomial and the sign of the norm of $y^2 - 4$, or from the quartic criterion. The columns of U are independent and S is nonsingular. The compressed interlacer $-(X+1)$ is primitive, giving height one. Finally

$$f_K P_S - f_S P_K = -\frac{1}{5}X^2(X-2)(11X-1),$$

which has only one factor $X-2$, contrary to the rank-one divisibility (12). □

Theorem 11.2 (Infinite exact rank-two Salem surgery)**.** *Let*

$$\nu \ge 3, \qquad \nu \equiv 3 \text{ or } 7 \pmod{10}, \qquad K = \nu^2 + 1, \qquad a = \frac{7\nu^2 - 3}{10}. \tag{65}$$

There exists a rational symmetric S_ν in the fixed plane (44) such that the update has rank exactly two and

$$f_\nu = X(X^2 - aX - K), \qquad P_\nu = (X-2)(X+1). \tag{66}$$

One branch satisfies

$$\det S_\nu = -\frac{3(\nu-1)^2K^2}{10\nu^2(\nu+1)^2} < 0. \tag{67}$$

The reciprocal polynomial

$$R_\nu(z) = z^4 - az^3 + (2-K)z^2 - az + 1 \tag{68}$$

is an irreducible Salem polynomial. Moreover

$$\kappa(B_K) = K = \nu^2 + 1 \longrightarrow \infty, \qquad \kappa(R_\nu) = 1, \tag{69}$$

and the same full–minor output cannot be produced by a rank-one update of the base pencil.

Proof. The congruence in (65) makes K, a integers. Substitution in the square class of Theorem 9.1 gives

$$K(K-1)(2K-5)(K-1-a)(2a+K-4) = \left(\frac{3}{5}K(2K-5)\nu\right)^2.$$

Hence a rational S_ν exists. Solving the determinant quadratic with one choice of square root gives (67); it is nonzero, so the update has rank two.

Let $q = X^2 - aX - K$. Direct evaluation gives

$$q(-2) = 2 + \frac{2}{5}K > 0, \qquad q(-1) = -\frac{3}{10}K < 0, \tag{70}$$

$$q(0) = -K < 0, \qquad q(2) = 6 - \frac{12}{5}K < 0. \tag{71}$$

Thus its roots satisfy $-2 < \alpha < -1 < 0 < 2 < y$. If q were reducible over $\mathbf{Q}$, its root in $(-2,-1)$ would be an integer, impossible. Finally

$$\mathrm{N}_{\mathbf{Q}(y)/\mathbf{Q}}(y^2-4) = q(2)q(-2) = -\frac{12}{25}(2K-5)(K+5) < 0. \tag{72}$$

A square in the real quadratic trace field has nonnegative norm. Hence y^2-4 is not a square, and the reciprocal lift is irreducible. The root placement proves the Salem signature.

The output interlacer $-(X+1)$ is primitive, so $\kappa(R_\nu) = 1$, whereas Proposition 5.2 gives the base height. Finally

$$f_K P_\nu - f_\nu P_K = \frac{1}{K}X^2(X-2)\big[(aK - K^2 + 2K - 1)X + a\big]. \tag{73}$$

At $X = 2$ the bracket equals $-\frac{3}{10}(2K-5)(K-2) \neq 0$. Thus $X-2$ occurs only once, violating the rank-one square divisibility. □

The first two instances are

ν	K	a	$R_\nu(z)$
3	10	6	$z^4 - 6z^3 - 8z^2 - 6z + 1$,
7	50	34	$z^4 - 34z^3 - 48z^2 - 34z + 1$.

12. Scope, novelty, and open directions

Theorems 3.1 and 6.2 classify the rational image of a *fixed cyclic realization* together with its fixed first principal compression. They are not claims about arbitrary matrices, arbitrary symmetrizers, or the full Jacobi inverse problem. Rank-m characteristic-polynomial assignment, restricted-rank eigenvalue modification, and algebraic relations among principal minors have substantial independent literatures; see [14, 2, 1]. Within that landscape, the feature isolated here is the paired full–compression assignment law for one cyclic self-adjoint model, including a converse, followed by exact arithmetic and Salem applications. We do not claim novelty for the secular, Rayleigh-square, local–global, elliptic-rank, or quartic-Salem ingredients individually; the new statements are their constrained simultaneous-surgery consequences.

The negative results close four sharply delimited routes. First, pointwise full and compressed eigenspace preservation imposes the rank bound of Theorem 7.2. Second, semidefinite perturbations meeting the two ordered output pins and strict integral Salem signature are excluded in every rank by Theorem 7.5. Third, the binary norm condition in Theorem 8.1 and its infinite local obstruction apply at every rank while the particular canonical form H_K is retained. Fourth, in the particular plane $(T_K e, T_K^2 e)$, the exact slice admits neither a vanishing integral exterior shift nor an infinite family with any fixed bounded shift. The rational-point determination used for the nearest shift is known for the elliptic curve $240a3$ [12]; the new content here is its derivation as the exact spectral feasibility curve and the resulting surgery obstruction.

The positive results are complementary. The $K = 5$ certificate and Theorem 11.2 show that the rank-one square obstruction is genuinely removed and that exact arithmetic height collapse occurs in rank two. Proposition 9.2 also shows that every strict branch in the complete canonical slice is forced to be indefinite, rather than this being an accident of the exhibited family.

None of these statements excludes a construction that changes H_K, uses a different rational two-plane, increases the ambient order, or gives up the paired eigenspace pins. Nor do the successful updates preserve tridiagonal Jacobi sparsity or give an output for every integer K. The infinite family has an exterior shift of order K, so it does not prove a Boyd closure conjecture. A closure-relevant construction would still have to combine higher order, simultaneous integrality, bounded or vanishing exterior shift, Salem signature, and irreducibility. These are genuine open directions, not consequences claimed in this paper.

Appendix A. The target-slice elimination

The following calculation makes the elimination in Theorem 9.1 reproducible without relying on a computer-algebra black box.

Lemma A.1 (Affine coefficient solution)**.** *Put* $\delta = (K-1)(2K-5)$ *and write* $S = \left(\begin{smallmatrix} r & s \\ s & u \end{smallmatrix}\right)$ *and* $d = \det S$. *Coefficient comparison in* (31) *with target*

(45) *first forces* $m = K$ *and then gives*

$$\begin{aligned} r &= \frac{aK^4 - 4aK^3 + 4aK^2 - 3aK - K^6 + 5K^5 - 9K^4 + 9K^3 - 5K^2 + K}{\delta} \\ &\quad + \frac{-K^4 + 5K^3 - 9K^2 + 8K - 3}{K}\, d, \\ s &= \frac{-2aK^3 + 4aK^2 - aK + 2K^5 - 8K^4 + 10K^3 - 6K^2 + 2K}{2\delta} \\ &\quad + \frac{2K^3 - 8K^2 + 9K - 4}{2K}\, d, \\ u &= \frac{aK^2 - K^4 + 3K^3 - 2K^2}{\delta} + \frac{-K^2 + 3K - 1}{K}\, d. \end{aligned}$$

After clearing denominators, let $\mathcal{Q}_{K,a}(d)$ *be the numerator of* $ru - s^2 - d$. *It is a quadratic with leading coefficient*

$$-(K-2)^2(K-1)^2(2K-5)^2$$

and discriminant

$$-16K^5(K-1)^3(2K-5)^3(a-K+1)(2a+K-4).$$

Consequently $\mathcal{Q}_{K,a}$ *has a rational root if and only if the square-class condition* (49) *holds.*

Proof. Insert $p_1 = X, p_2 = X^2$ in the remainder formulas of Lemma 2.2, polarize to obtain C and D, and compare the coefficients of $f_S - f_{a,m}$ and $P_S - P_*$. Gaussian elimination over $\mathbf{Q}(K, a, d)$ gives the three displayed affine expressions and $m = K$. Substitution in $ru - s^2 = d$ gives $\mathcal{Q}_{K,a}$; expanding its leading coefficient and its discriminant gives the displayed factors. Its leading coefficient is nonzero for $K \geq 5$. Finally its discriminant equals

$$\big(4K^2(K-1)(2K-5)\big)^2 K(K-1)(2K-5)(K-1-a)(2a+K-4),$$

which proves the last assertion, including the zero-discriminant case. □

Appendix B. Explicit matrix certificate for the infinite family

For completeness we display the branch used in Theorem 11.2. Put $K = \nu^2 + 1$ and $D = 10\nu^2(\nu+1)^2$. Define

$$\begin{aligned} U_{11} &= 2\nu^{10} + 16\nu^9 + 9\nu^8 + 2\nu^7 + 13\nu^6 + 12\nu^5 + 10\nu^4 - 4\nu^3 + 6\nu^2 + 4\nu + 2, \\ U_{12} &= 2\nu^8 + 16\nu^7 + 9\nu^6 + 2\nu^5 + 11\nu^4 + 2\nu^3 + 6\nu^2 - 2\nu + 2, \\ U_{22} &= 2\nu^6 + 16\nu^5 + 9\nu^4 + 2\nu^3 + 9\nu^2 - 8\nu + 2. \end{aligned}$$

Then one may take

$$S_\nu = \frac{K}{D}\begin{pmatrix} -U_{11} & U_{12} \\ U_{12} & -U_{22} \end{pmatrix}. \tag{74}$$

Direct expansion gives (67) and the output pair (66).

An entirely rational base realization is also explicit. With $\mathcal{B}_K, b_1, b_2$ as in (39), set

$$a_0 = \frac{K^2 - 3K + 1}{K}, \qquad a_1 = \frac{4K^4 - 12K^3 + 6K^2 - 3K + 1}{K\mathcal{B}_K},$$
$$a_2 = -\frac{2(K^2 - 3K + 1)}{\mathcal{B}_K}.$$

With

$$T_K = \begin{pmatrix} a_0 & 1 & 0 \\ b_1 & a_1 & 1 \\ 0 & b_2 & a_2 \end{pmatrix}, \qquad H_K = \operatorname{diag}\left(1, b_1^{-1}, (b_1 b_2)^{-1}\right),$$

one has $H_K T_K = T_K^{\mathsf{T}} H_K$, the characteristic pair is (f_K, P_K), and the update plane is exactly $U_K = (T_K e, T_K^2 e)$. Substitution of (74) verifies the two target characteristic polynomials over $\mathbf{Q}(\nu)$.

Appendix C. Exact certificates

All displayed elimination identities and the two explicit Salem surgeries were verified over the rational function field with exact symbolic arithmetic. The companion scripts included with the source package check:

(1) the rank-one coefficient elimination and all $K = 5, \ldots, 150$ against an enlarged integral search window;
(2) the cofactor and Gram–Plücker identities;
(3) the $K = 5$ rational H-self-adjoint matrix realization;
(4) the target-slice eliminant and discriminant;
(5) the same-form norm identity and the first local obstruction instance;
(6) the parametrized slice conic, determinant eliminant, and forced negative determinant;
(7) the birational near-shadow identity, the displayed rational points, and the point counts over $\mathbf{F}_7$ and $\mathbf{F}_{11}$;
(8) and the infinite family, including (67), (73), rank, and reciprocal output.

These computations certify displayed algebra; they do not replace the proofs of the converse or the root-location arguments.

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